North East University Bangladesh

PROJECT PROPOSAL

Lung Cancer Detection from CT Scan Images Using Convolutional Neural Network

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Abstract

Lung cancer is one of the leading causes of cancer related death worldwide, and early detection is essential for improving patient survival and treatment planning. Computed Tomography (CT) scan images provide detailed anatomical information about lung tissues and are widely used for detecting nodules and abnormal growth. However, manual interpretation of CT images can be time consuming and may be affected by human error, especially when the abnormal region is small or visually similar to normal lung tissue. This project proposes a computer aided diagnosis system for lung cancer detection using a Convolutional Neural Network (CNN). The proposed system follows a structured workflow that includes importing libraries and dataset, data visualization, image data processing, model development, and model evaluation. The CNN architecture will include three convolutional layers with max pooling, two fully connected layers, and a softmax output layer for classifying CT images into normal, benign, and malignant categories. The model will be evaluated using accuracy, loss curves, confusion matrix, precision, recall, F1-score, and classification report. The expected outcome is a simple, understandable, and effective deep learning based system that can support radiologists by providing an automated second opinion for early lung cancer detection.

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List of Symbols and Acronyms

Acronym	Meaning
AI	Artificial Intelligence
AUC	Area Under the Curve
CAD	Computer Aided Diagnosis
CNN	Convolutional Neural Network
CT	Computed Tomography
DL	Deep Learning
FCM	Fuzzy C-Means
FN	False Negative
FP	False Positive
IQ-OTH/NCCD	Iraq-Oncology Teaching Hospital/National Center for Cancer Diseases dataset
ML	Machine Learning
ReLU	Rectified Linear Unit
TN	True Negative
TP	True Positive
XAI	Explainable Artificial Intelligence

1. Introduction

1.1 Background

Lung cancer is a severe medical condition caused by abnormal and uncontrolled growth of cells in lung tissue. It may appear as small nodules, masses, or abnormal tissue regions in medical images. Early diagnosis is very important because treatment is usually more effective when cancer is detected before it spreads to other organs. Among different imaging techniques, CT scan imaging is highly useful because it provides detailed cross sectional views of the lungs and helps clinicians identify nodules, lesion boundaries, and suspicious tissue patterns.

In recent years, Artificial Intelligence and Deep Learning have become important tools in medical image analysis. Convolutional Neural Networks are especially suitable for image classification because they can learn edges, textures, shapes, and deeper visual features directly from images. In lung cancer detection, CNN models can analyze CT scan images and classify them into normal, benign, and malignant categories. Such systems can support radiologists, reduce workload, and improve diagnostic consistency.

1.2 Problem Statement

Manual examination of lung CT scan images requires expert knowledge and careful visual inspection. Small nodules, low contrast regions, and complex lung structures can make diagnosis difficult. In addition, manual interpretation may be slow and subjective when many images need to be examined. Although advanced CNN models have shown high performance, some models are complex, computationally expensive, and difficult to interpret. Many medical datasets are also limited or imbalanced, which can cause overfitting and reduce performance on unseen images.

Therefore, this project proposes a simple CNN based lung cancer detection system that follows a clear process flow from dataset preparation to model evaluation. The goal is to build a practical academic model that can classify lung CT scan images and evaluate performance using reliable metrics.

1.3 Aim of the Project

The main aim of this project is to design, train, and evaluate a Convolutional Neural Network based system for detecting lung cancer from CT scan images.

1.4 Objectives

- To collect or prepare a lung CT scan image dataset for classification.
- To visualize CT scan images and examine class distribution before training.
- To preprocess CT images using resizing, normalization, and augmentation.
- To develop a simple CNN model with three convolutional layers, max pooling, two fully connected layers, and a softmax output layer.

- To train the model using training data and validate it using validation data.
- To evaluate the model using accuracy, loss curve, confusion matrix, precision, recall, F1-score, and classification report.
- To present the system as a supportive computer aided diagnosis tool rather than a replacement for medical experts.

1.5 Research Questions

- Can a simple CNN model classify lung CT scan images accurately?
- Which preprocessing steps help improve CNN performance for lung cancer detection?
- How well does the trained model perform on validation and testing data?
- Can confusion matrix and classification report identify class wise model strengths and weaknesses?

2. Literature Review

Pandian et al. proposed a lung cancer detection and classification method using CNN, GoogleNet, and VGG16 based architectures on CT scan images. Their study emphasized that automated tools can reduce misdiagnosis and support early detection. The study reported high classification performance and showed that CNN based image analysis can distinguish normal and malignant lung tissues effectively [1].

Jozi and Al-Suhail introduced LCxNet, an explainable CNN framework for lung cancer detection using CT images. Their work used the IQ-OTH/NCCD dataset, which contains normal, benign, and malignant classes. The study compared multiple optimizers, including SGD, RMSProp, Adam, AdamW, and NAdam. It also used Grad-CAM and t-SNE to improve interpretability and clinical trust. This is important because medical AI systems should not only provide predictions but also help users understand which image regions influenced the decision [2].

Shariff, Paritala, and Ankala proposed a CNN model with Differential Augmentation for non-small cell lung cancer detection. Their work highlighted memory overfitting as a major problem in medical image classification. By applying augmentation strategies such as brightness, hue, saturation, and contrast changes, they improved model generalization and achieved strong performance. This supports the use of image augmentation in the proposed project [3].

Shah et al. proposed a deep ensemble 2D CNN approach for lung nodule detection using the LUNA16 dataset. Instead of depending on a single CNN, the study combined multiple CNN models to improve prediction accuracy. The study showed that ensemble learning can improve classification performance, but it may increase computational complexity. For this project, a simpler CNN is selected to make the workflow easier to implement and explain [4].

Prakashbabu, Ashok Kumar, and Vithyaa proposed a lung cancer detection model using CNN and Fuzzy C-Means clustering. Their work emphasized image preprocessing, denoising, segmentation, and feature extraction. This supports the idea that proper preprocessing is essential before training a CNN model [5].

2.1 Summary of Related Studies

Table 1: Summary of selected related studies

Author/Year	Method	Dataset/Images	Key Idea	Limitation
Pandian et al. (2022)	CNN, GoogleNet, VGG16	CT scan images	Deep learning based lung cancer detection and classification	Advanced architecture may require more computation
Jozi and Al-Suhail (2025)	LCxNet with XAI	IQ-OTH/NCCD	Normal, benign, and malignant classification with Grad-CAM	Needs clinical validation before real deployment
Shariff et al. (2025)	CNN + Differential Augmentation	Multiple CT datasets	Augmentation to reduce memory overfitting	Generalization still requires wider clinical testing
Shah et al. (2023)	Deep ensemble 2D CNN	LUNA16	Combines multiple CNNs for better accuracy	Higher complexity than a single CNN
Prakashbabu et al. (2019)	CNN + Fuzzy C-Means	CT, MRI, ultrasound	Preprocessing and segmentation based detection	Older model and limited deep learning evaluation

2.2 Research Gap

The reviewed studies show that CNN based lung cancer detection can achieve good performance. However, several challenges remain. Some models are too complex for beginner level implementation, some have limited interpretability, and many studies face overfitting because of small or imbalanced datasets. This project addresses these issues by proposing a clear and simple CNN workflow with proper preprocessing, validation, and model evaluation. The project focuses on practical implementation and understandable reporting rather than only achieving high accuracy.

3. Proposed Methodology

The proposed methodology follows the process flow shown in Figure 1. The workflow begins with importing libraries and the dataset. Then sample images and class distribution are visualized. After that, image data processing is performed to prepare the CT images for CNN training. The processed data are divided into training and validation data. The CNN model is then developed, trained, and evaluated using graphical and numerical performance metrics.

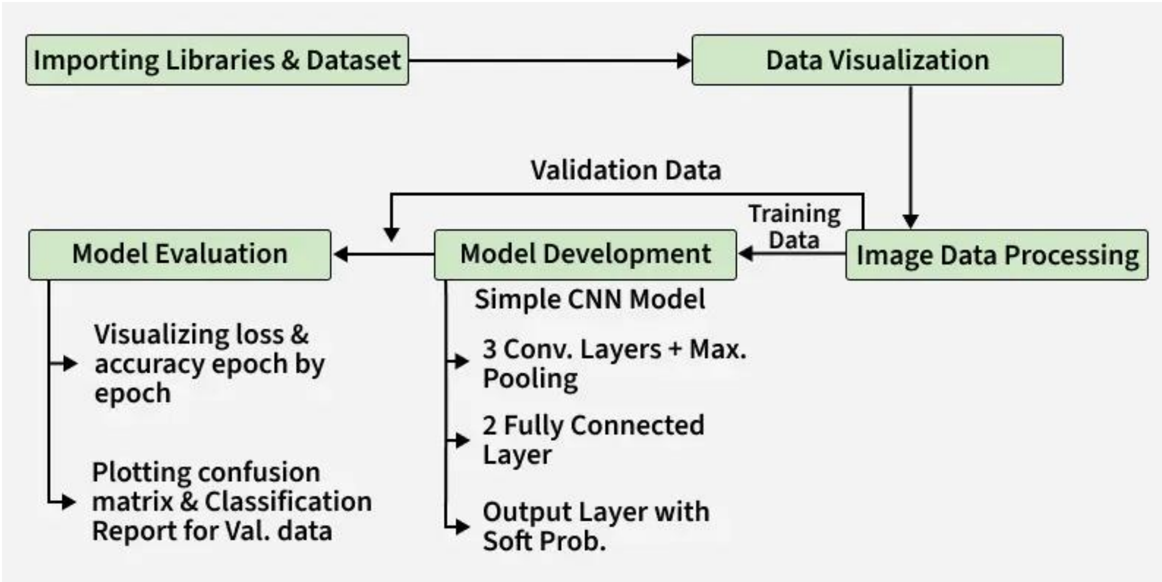

Figure 1: Proposed process flow for lung cancer detection using CNN

3.1 Dataset Collection

The project will use lung CT scan image data. The preferred dataset is the IQ-OTH/NCCD lung cancer dataset because recent studies used it for three class classification into normal, benign, and malignant categories. If the dataset is not available, another publicly available CT scan dataset such as LUNA16 may be used. The images will be organized into separate folders according to class labels.

3.2 Data Visualization

Data visualization will be performed before model training. Sample images from each class will be displayed to understand visual patterns. Class distribution will also be plotted to check whether the dataset is balanced. This step is important because class imbalance can make the model biased toward the majority class.

3.3 Image Data Processing

All CT scan images will be resized to a fixed dimension such as 224 x 224 pixels or 128 x 128 pixels. Pixel values will be normalized between 0 and 1. Images may be converted to grayscale or kept in RGB format depending on the dataset structure and model input requirement. Augmentation methods such as rotation, zooming, shifting, brightness adjustment, and contrast adjustment will be applied to the training images to reduce overfitting and improve generalization.

3.4 Model Development

The proposed model is a simple CNN designed for image classification. It contains three convolutional layers with max pooling, followed by two fully connected layers and a softmax output layer. ReLU activation will be used in hidden layers because it helps the model learn nonlinear image features efficiently. Dropout may be added to reduce overfitting.

Table 2: Proposed CNN architecture

Layer	Description	Purpose
Input Layer	Receives resized CT scan images	Provides image data to the CNN
Convolution Layer 1 + Max Pooling	Extracts basic edges and textures	Learns low level image features
Convolution Layer 2 + Max Pooling	Extracts deeper lung tissue patterns	Learns middle level features
Convolution Layer 3 + Max Pooling	Extracts complex abnormal patterns	Learns high level features
Flatten Layer	Converts feature maps into vector form	Prepares features for dense layers
Fully Connected Layer 1	Dense decision layer	Learns feature combinations
Fully Connected Layer 2	Dense decision layer	Improves classification decision
Softmax Output Layer	Outputs class probabilities	Predicts normal, benign, or malignant class

3.5 Training and Validation

The dataset will be divided into training, validation, and testing sets. A possible split is 70 percent training, 15 percent validation, and 15 percent testing. The training set will be used to train the CNN model, the validation set will be used to monitor learning during epochs, and the test set will be used for final performance evaluation.

3.6 Model Evaluation

The model will be evaluated using both visual and numerical methods. Training and validation accuracy will be plotted epoch by epoch. Training and validation loss will also be plotted to detect overfitting. A confusion matrix will be used to show correct and incorrect predictions for each class. A classification report will provide precision, recall, F1-score, and support. Recall will be considered very important because missing a malignant case may be clinically risky.

4. Expected Results and Evaluation

The expected result is a CNN based lung cancer detection model that can classify CT scan images into normal, benign, and malignant categories. The model is expected to show good accuracy if the dataset is properly cleaned, balanced, and augmented. The project will also generate loss and accuracy graphs, a confusion matrix, and a classification report. These outputs will help explain whether the model is learning correctly and whether any class is being misclassified frequently.

The final model should not be considered a complete clinical diagnostic system. Instead, it will be presented as an academic prototype and a supportive computer aided diagnosis tool. The model result can help guide medical image review, but final diagnosis must be made by qualified medical professionals.

4.1 Evaluation Metrics

- Accuracy will measure the overall proportion of correct predictions.
- Precision will measure how many predicted positive cases are actually correct.
- Recall will measure how many actual positive cases are correctly detected.
- F1-score will balance precision and recall.
- Confusion matrix will show detailed class wise prediction performance.
- Loss and accuracy curves will show training behavior across epochs.

5. Project Plan

Table 3: Proposed project timeline

Week	Activity
Week 1	Literature review and dataset selection
Week 2	Dataset collection and folder organization
Week 3	Data visualization and class distribution analysis
Week 4	Image preprocessing and augmentation
Week 5	CNN model design and compilation
Week 6	Model training and validation
Week 7	Model testing and evaluation using confusion matrix and classification report
Week 8	Final report writing and presentation preparation

5.1 Required Tools and Technologies

- Python programming language
- Google Colab or Jupyter Notebook
- TensorFlow/Keras for CNN implementation
- OpenCV or PIL for image processing
- NumPy and Pandas for data handling
- Matplotlib and Seaborn for visualization
- Scikit-learn for confusion matrix and classification report

6. Conclusion

This project proposal presents a CNN based lung cancer detection system using CT scan images. The proposed workflow follows the provided process flow, including importing libraries and dataset, data visualization, image data processing, model development, and model evaluation. The CNN model will use three convolutional layers with max pooling, two fully connected layers, and a softmax output layer. The system will be evaluated using accuracy, loss curve, confusion matrix, and classification report. The project is expected to provide a clear, simple, and effective academic prototype for lung cancer detection using deep learning. It may support radiologists by providing an automated second opinion, but it will not replace professional medical diagnosis.

Appendix

Appendix A: Proposed Process Flow

The proposed process flow starts from importing libraries and dataset. The data are visualized to understand image classes and distribution. The CT scan images are then processed through resizing, normalization, and augmentation. The processed images are divided into training and validation data. A simple CNN model is developed and trained. Finally, the model is evaluated using loss and accuracy graphs, confusion matrix, and classification report.

Appendix B: Ethical Considerations

Medical image data must be handled carefully. Any dataset used in this project should be publicly available, anonymized, and used only for academic research. The project will not use personal patient identity information. The proposed model will be presented only as a supportive research prototype.

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